

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2019

vii) If $Q = 1$, the output is said to be

- a) Reset
- b) Set
- c) Previous state
- d) current state

viii) Which one of the following are also called ripple counters?

- a) SSI counters
- b) Synchronous counters
- c) Asynchronous counters
- d) VLSI counters

ix) How many flip-flops are required to construct MOD-30 counter?

- a) 5
- b) 6
- c) 4
- d) 8

x) How much storage capacity does each stage in a shift register represent?

- a) One bit
- b) Two bits
- c) Four bits
- d) Eight bits



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2019

Bachelor in Computer Applications
Course Title: Digital Logic
Code No: CACS 105
Semester: 1st

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Subtract: $1010.110 - 101.101$ using both 2's and 1's complement. [5]
3. Simplify (Using k-map) the given Boolean function in both SOP and POS using the don't care condition d:

$$F(A, B, C, D) = \pi(0, 1, 3, 7, 8, 12) \text{ and } \pi d(5, 10, 13, 14) \quad [2 + 3]$$

4. Define decoder. Draw logic diagram and truth table of 3 to 8-line decoder. [1 + 4]
5. Define ROM. Implement the following combinational logic function using ROM: [2 + 3]

A1	A0	F1	F2
0	0	1	0
0	1	0	1
1	0	1	1
1	1	1	0

6. What are the drawbacks of clocked RS flip flop? Explain the operation of JK Flip flop along with its circuit diagram and characteristic table. [2+3]
7. What is T flip-flop? Explain clocked JK flip-flop with its logic diagram and truth table. [1 + 4]

8. Design MOD - 7 counter with state and timing diagram.

[2 + 1 + 2]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Define PLA. Design a PLA circuit with given functions.

$$F1(A, B, C) = \Sigma(3, 5, 6, 7)$$

$$F2(A, B, C) = \Sigma(0, 2, 4, 7). \text{ Design PLA program table also.}$$

[3 + 7]

10. Distinguish between sequential and combinational logic with example? Discuss the design procedure of combinational logic.

[4+6]

11. A sequential circuit with two D flip-flops, A and B, two inputs x and y, and one output z, is specified by the following next state and output equations

[4+3+3]

$$A(t+1) = x'y + xA$$

$$B(t+1) = x'B + xA$$

$$z = B$$

a) Draw the logic diagram.

b) Derive the state table.

c) Derive the state diagram.